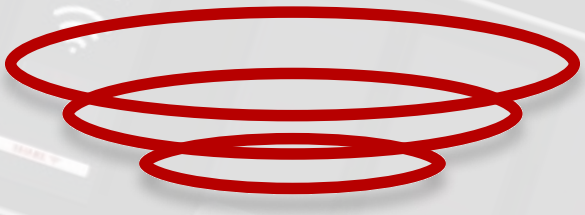
WIRECAST



FRANCESCO RASTELLI
UI/UX

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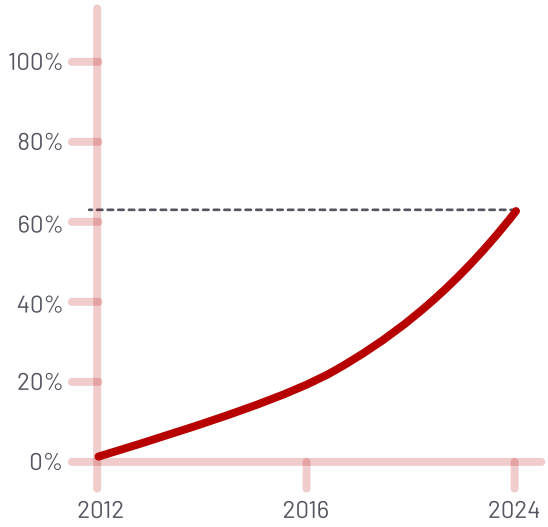
REFERENCES 18

Word count: 3298

THE PROBLEM 2

ABSTRACT

Homes are becoming increasingly smarter each year. The first true smart thermostat was introduced by **Ecobee** in 2007 [1], followed by **Belkin's** WeMo Insight Switch (2012) [2] and the **Amazon Echo** in 2014 [3], revolutionising home automation and energy management. Growing eco-consciousness has led to a sharp rise in smart energy meter installations—from just **1%** of homes in 2012 to over **64%** in the UK by 2024 [4]. However, these systems focus on overall household power management, neglecting individual devices that waste energy when left on or used inefficiently. This project addresses that gap by integrating smart plugs for optimal control.



USER PERCEPTIONS AND INTENTIONS

The residential sector accounts for **22%** of global energy consumption and **17%** of CO_2 emissions [5]. To reduce this impact and help users manage rising energy costs, smart home energy management systems **(SHEMS)** must be widely adopted. Effective implementation should focus on three key areas:

- Providing **control** over both household and individual device energy usage.
- **Integrating** smart plugs with a clear, user-friendly interface.
- Connecting to **smart grids** to optimise energy use with dynamic pricing.

MOTIVATIONS [5]

15-23% saved on energy bills annually

25.2% adopted smart tech due to intrest

24% sought increased comfort

16.4% sought sustainability

BARRIERS [5]

59% did not use a SHEMS

50.1% noticed cost savings

49% reported a decrease in energy use

DEALING WITH DATA SAFETY

One of the key concerns that users have when attempting to adopt SHEMS is "**data security and privacy**, which is most often stated in the existing literature as a major barrier to adoption" [5]. While this project does not go into a lot of detail around how this problem could be tackled, it is still something that I feel should be mentioned as a key reason around why there are few adopters of smart plugs at the moment. This can be especially seen with research finding **13** vulnerabilities in only **9** plugs. [6]

THE PROBLEM

EXISTING RESEARCH

When looking at peer-reviewed papers, I found multiple research around the integration of SHEMS in the house. One that provided excellent insights was focused on user-centred design and evaluation, which led to multiple insights [7]. Target users were around **18-30** years old and included people familiar with both smart technology and ones who'd just been introduced to it:

PAIN POINTS

- Understanding what devices **consume** the most energy.
- Lack of **real-time feedback.**
- Fragmented **control** of smart devices around the house.
- **Privacy** concerns around data collected by the plugs and being stolen.

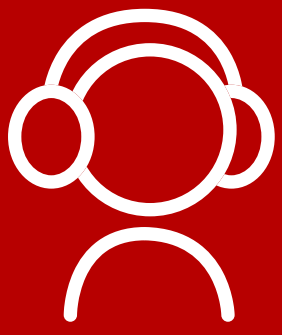
RESEARCH METHODOLOGY

- A questionnaire to collect demographic data was run around current experience and expectations.
- Interviews were run to find out about their energy consumption habits, awareness of costs and current management methods.

KEY INSIGHTS

- Tech-savvy users were used to smart thermostats and smart plugs, but found their UI's **confusing**. They also wanted better integration with dynamic **pricing**.
- New users struggled to **track costs** and were interested in energy saving features but struggles to **set them up**.

USER PERSONA



TECH-SAVY



25-40 YEARS OLD



HAS HIGH ENERGY BILLS



ECO-CONSCIOUS

EXISTING SOLUTIONS 4

TASMOTA SMART PLUG

As a smart plug, the device acts as an extension, not replacing the socket, ensuring easier installation. Looking at the 5 key criteria, some issues arise.

- **Visibility**: Contrast in buttons is visible, although text indicating usage is very small and does not stand out.
- **Affordance**: Buttons are a different colour to text.
- **Mapping**: Everything is squished together and badly spaced.
- **Constraints**: No constraints due to lack of possible actions.
- **Feedback**: No feedback except text changing from OFF to ON.

When looking at user reviews, a major issue was the requirement for the plugs to be calibrated, which often didn't work or users were unable to do as they were missing the proper equipment. [8]

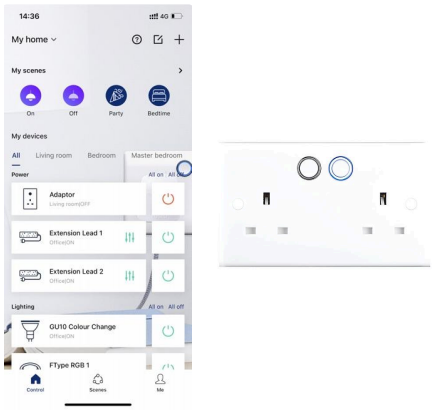


BG SMART SOCKET

Smart socket replace an outlet and are installed directly into a wall. To control them, you are expected to connect them to wifi.

- **Visibility**: Some actions like power are clear, although buttons like "scenes" don't indicate clearly what they do.
- **Affordance**: Power buttons are clear, although colour blind people would struggle without text.
- **Mapping**: Most mapping is unclear.
- **Constraints**: No constraints due to lack of possible actions.
- **Feedback**: Visible feedback in terms of colour change, but power usage is unclear.

This is another product which while it sometimes works, reviews often state it being a hassle to set up with it failing multiple times leading to them having to control it manually. [9]



EVE ENERGY OUTLET

This is rated as one of the best, connecting with home control apps, although you need to install their app to control power.

- **Visibility**: Some contrast is visible, although the app is very cluttered. Graph is quite unclear.
- **Affordance**: Buttons are clearly visible.
- **Mapping**: Everything is squished together and badly spaced.
- **Constraints**: No constraints due to lack of possible actions.
- **Feedback**: No feedback except button colour switching from ON to OFF

As stated above, reviews have mostly been positive. [10]



USER RESEARCH 5

User research was focused on conducting user interviews due to time constraints resulting in a lack of time for a survey to be able to explore a wide enough demographic to get useful and actionable insights from. Interviews were conducted with **3 people** who ranged from very tech-savvy, to having no smart technology in the home, in order to be able to assess the integration of the system across a range of people. These can be viewed at [19].

KEY QUESTIONS

7 key questions needed to be answered in my interviews to be able to get an accurate understanding of the way people currently use smart technology in the home, reasons as to why they may not use it, and things like pain and pleasure points, all to lead to key insights in order to create a viable USP. The questions asked were:

1. Do you currently use any smart technology in the house; if so, what is it?
2. Why do you originally adapt smart technology into your house? If you haven't, why not?
3. Take me through the workflow of a smart technology control app you use?
4. Are there are pain or pleasure points that arise from this?
5. How do you currently manage your energy usage at home?
6. Are you ever worried about your energy bills? If so, have you done anything to lower them?
7. Have you heard or considered the introduction of smart plugs in the house?

KEY POINTS



TOTAL CONTROL



HIGH BILLS



FRAGMENTED CONTROL



ECO-CONSCIOUSNESS



MONITORING



MANAGEMENT



AUTOMATION



CLEAR BENEFITS



VISUALISATION

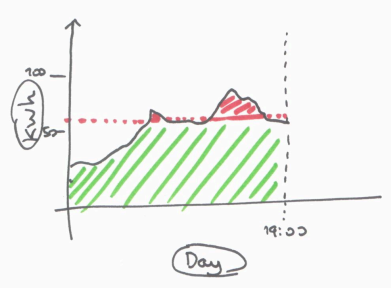
UNIQUE SELLING POINT 6

ENERGY MONITORING

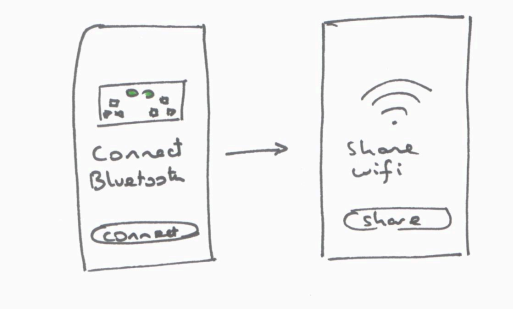
Users require easy to understand energy monitoring of both the household as a whole and of individual devices.

- Key information is current energy usage and history of energy usage to compare to.
- Values in kWH aren't useful to the everyday user unless you have something to compare it to or a colour scale.
- Day/week/month/year scale for history usage.





EASE OF SETUP



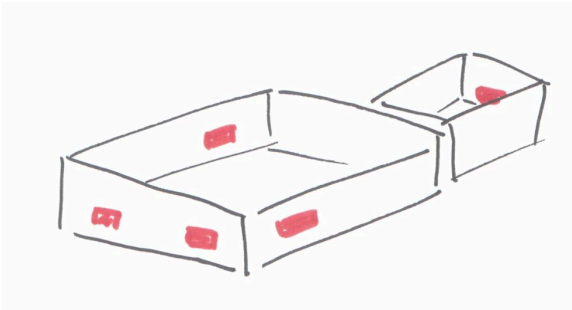
All of the existing products have issues with how complicated they are to set up. The proposed system needs to completely overturn this.

- What is the easiest way to connect multiple devices?
- Connect to one via Bluetooth and share wifi password with it.
- This then relays to the other plugs and connects them all.

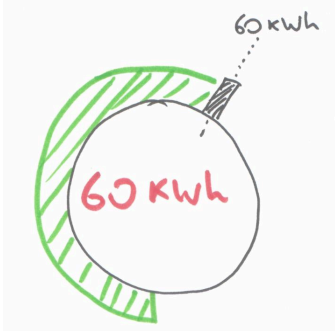
ENERGY MANAGEMENT

Controlling what devices are on and how much power they use is essential to the success of this app.

- Each plug has a specific device attached to it.
- Users can change the name and identify it that way.
- Plugs can work out distance between each other and use a ML algorithm to create a 3D model of the house.



COST MANAGEMENT



Controlling devices to run at periods of time in which the dynamic pricing is lowest, and how much energy devices consume.

- Using a rotary scale, the user can put a limit to energy usage / cost of a device.
- This can be reflected on the graph.
- The same scale can be used to control household energy usage and cost limits.

INITIAL SITE MAP

SECTION 1 - SET-UP

Home screen telling user to set up the product

Setting up your device

Instructions on how to connect to intial plug

Connect to Bluetooth

Instructions to share wifi password to set-up

Share Wifi Password

Automatic set up of the 3D model

Setting up Model

Graphic of model being generated shown to increase **perceived performance**

If user presses question mark, they will be brought to help page on website

SECTION 2 - HOME PAGE

Interactive model with information below it

Home Page with 3D Model

Scroll down

Shows House Data

Click on room

Shows 3D model of room

Scroll down

Shows Room Data

Click on plug

Shows Plug Information

Scroll down

Shows Plug Data

Clicking on the name allows user to change it

Click on search

SECTION 3 - SEARCH TAB

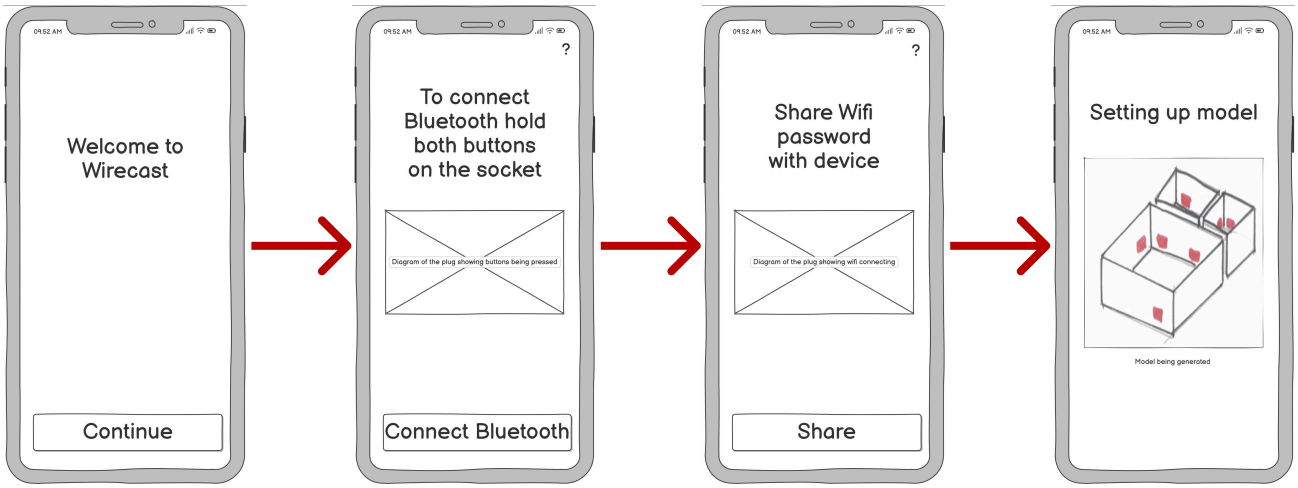
Search through all the rooms or plugs by name

List of all rooms / plugs

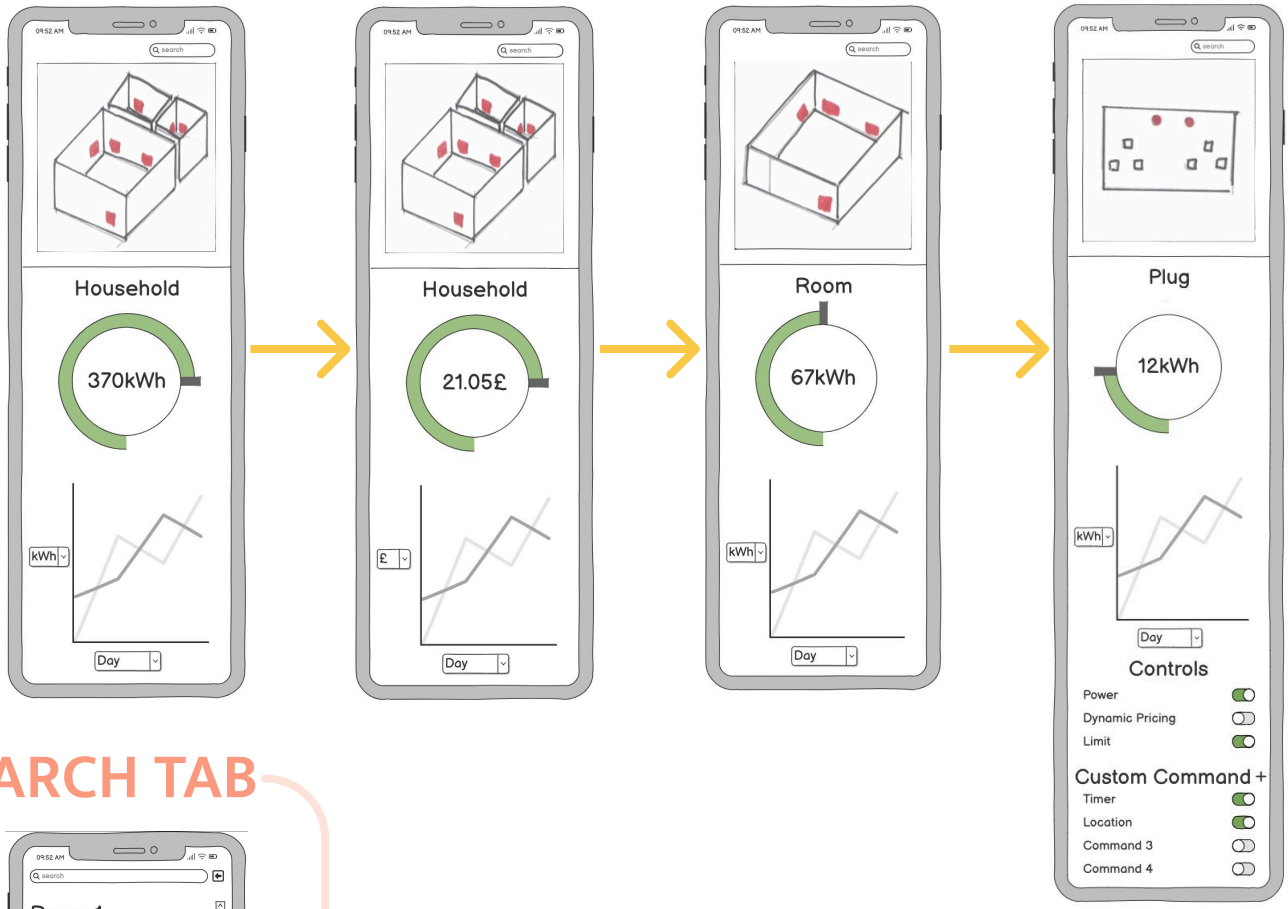
Click on plug / room

Shows Data

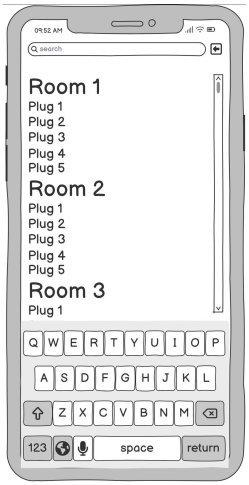
SET-UP



HOME PAGE



SEARCH TAB



An initial sketch up of the app was built in Balsamiq to be able to quickly test ideas with the users. It follows the same site map as the previous page, giving users the ability to interact and give feedback on workflow. Every scale of the model had it's own variations of both usage and cost. The results of these tests can be seen on the following page.

An interactive version of this app is available at [11]:
https://balsamiq.cloud/slqx1u7/p49899k/r2278?f=N4IgUiBcAMA0IDkpxAYWfAMhkAhHAsjgFo4DSUA2gLoC+QA=

USER TESTING

Users were tested against the sketch-up to try and work out:

1. **How they experienced the workflow of the product, with focus on:**
   a. Key sections they got stuck at.
   b. Site map structure and flow,
2. **Key features the app was missing.**
3. **Clarity of information layout on each page.**

1 WORKFLOW

- Often when reaching the plug, it wasn't intuitively obvious that they needed to press on it to go back to the room.
- The search bar was deemed very useful when trying to get to the plug
- Changing the scale by having to press on the values wasn't obvious.

2 GAPS

- The lack of back arrows made the process confusing to use
- The users would have liked to have a saving / normal mode
- Ability to group plugs together outside of just rooms (eg. for multiple heaters)

3 CLARITY

- The limits weren't obvious to the user and how you change them
- Would prefer the graph as a bar chart instead of line graph
- Controls were a bit difficult to find, would be better to have them as a separate page & add controls for rooms and household.

KEY ADDITIONS



BACK ARROWS



FUTURE COST ESTIMATION



BIGGEST ENERGY USERS



ADD CUSTOM PLUG GROUPS



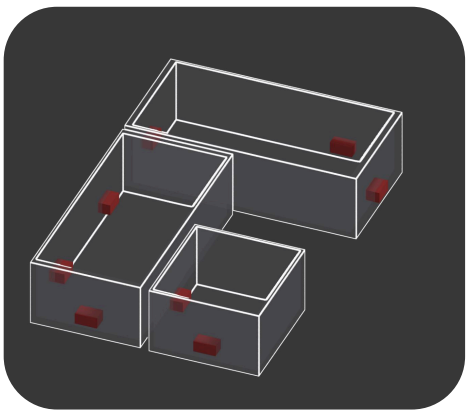
CONTROLS FOR ROOMS & HOUSEHOLDS



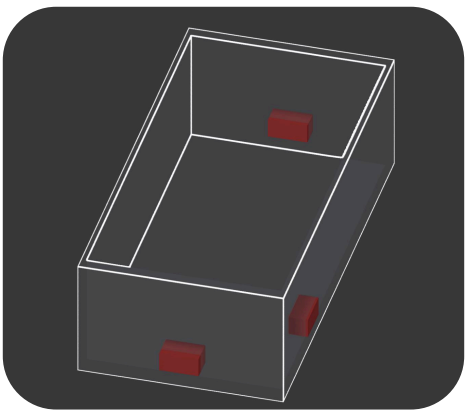
TOGGLE BETWEEN A SAVINGS/ NORMAL MODE

SPLICE MODEL 10

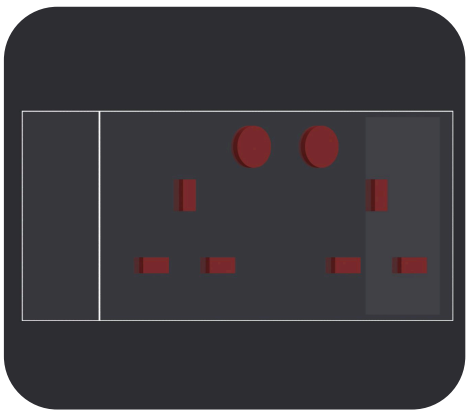
A 3D splice model was created to test the user interactions with the 3D model of the product. The 3D model is fully interactive, with the 3 scales being shows as household, room and finally plug.



Household shows all the rooms as it rotates. Hovering over a room moves all the other ones out of view and zooms into that room to access the respective plugs.



The room allows a user to click directly on the plug he wants to gain insights on, or add controls to the room directly.



When a plug is clicked on, it zooms into a view of it, allowing a user to see all its insights. Clicking back on it brings you back to room view.

To test the 3D Model yourself, as it's not implementable into Figma, please visit [12]:
https://my.spline.design/uiuxexperimentscopy-c385c9d45fdc1818858283f143353e5c/

USER FEEDBACK

FROM HOVER TO CLICK

DISABLE ORBIT

COLOUR CODING

COLOUR GRADING

PLAYFUL

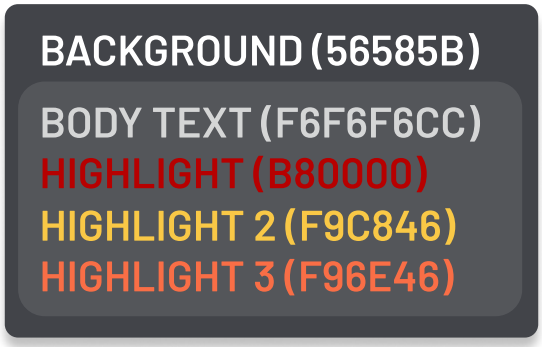
SMOOTH TRANSITIONS

COLOURS & TYPOGRAPHY

All text should be able to meet legibility standards set forward by the Web Content Accessibility Guidelines (WCAG 2.0) [13]. An easy to understand breakdown of this was given by Material Design [14], and followed when evaluating different colour choices for our design:

CONTRAST

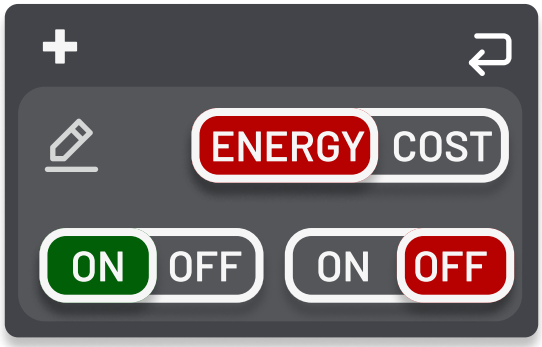
WCAG recommends a colour contrast of **"4.5:1 color contrast between text and background for normal text, and 3:1 to large text"** [14]. Evaluating our design, using WebAims Contrast Checker [15], gives us an AAA pass on text, but not on highlights.



Contrast Score:
4.91:1 - PASS
1.03:1 - FAIL
4.54:1 - PASS
2.49:1 - FAIL

ICONS

Icons and other elements don't need to meet WCAG legibility standards, but should still be **as visible as possible for clarity** [14]. As such, similar contrasts were maintained, with colour being used to highlight active and inactive states, interactive icons, and key information.

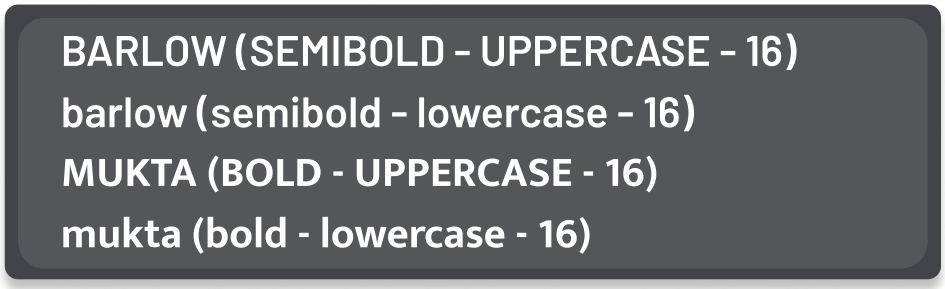


Opacity and visibility:
Active icons have a visibility score above 90%, with active and unfocused being around 70%. Change in colour such as the ON/OFF switch further improve visibility.

TYPOGRAPHY

Experimentation around the typography can be split into two parts. On one side, I experimented with the font and colour of the logo and title, eventually deciding on using **Syne Extra Bold at B80000**. On the other hand, I looked at the overall text, experimenting with both colour, fonts and casing. As there is little hierarchy required within the product, the same font and casing was used throughout the text for clarity, this ended up being **Barlow SemiBold (uppercase).**





ICONOGRAPHY

Systems icons should be "designed to be simple, modern, friendly, and sometimes quirky" [16]. Following Material Design's guidelines, icons were designed to utilise graphic, bold shapes, avoiding more thin strokes and organic designs. System icons were placed at a size of **24px** for optimal visibility without overpowering the user. Consistent stroke weights were used to ensure a feeling of even weight.


BACK


ENERGY USAGE


ADD COMMAND


TOP USERS


EDIT NAME


COMMANDS


SEARCH


TIMER

USER FEEDBACK


INCREASE TEXT CONTRAST


EVEN-OUT THE ICON LOOKS


POSITIVE COLOUR RESPONSE


CHANGE HIGHLIGHT COLOUR


STOP ICONS RESETTING


EASE OF SET UP

SET-UP

Welcome page is opened at the start to show the easy set-up procedure.

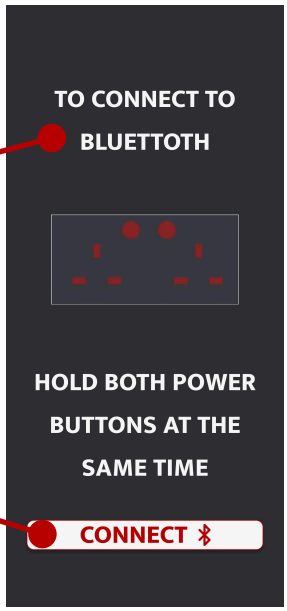


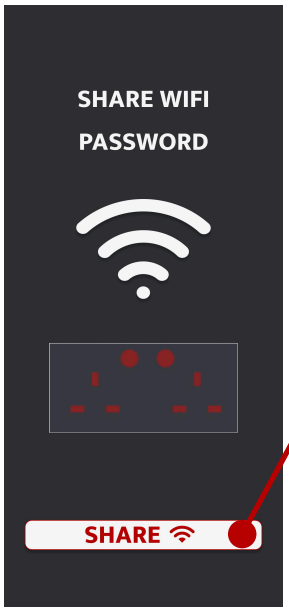


The logo is animated to bounce up and down.

Pressing continue brings you to the bluetooth set-up page.

Information on how to set up bluetooth on device.

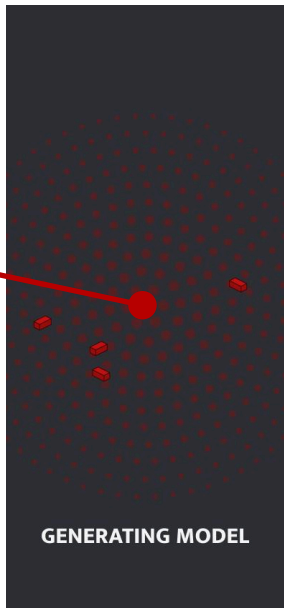


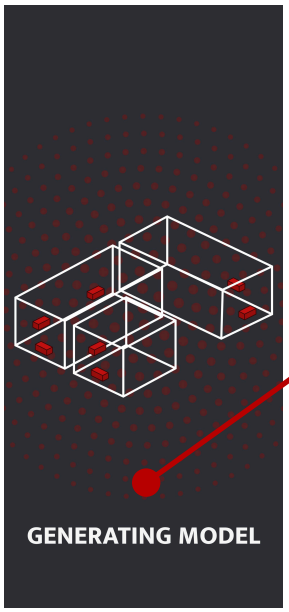


Wifi password can then be shared directly to it, where it then connects to all the other plugs in the house and begins the set up.

Pressing connect would bring up the overlay of available devices to connect to with bluetooth.

Set-up plays an animation of the 3D model being generated to improve the **perceived performance.**





When the model is fully created, the app automatically brings up the homepage.

HOMEPAGE

The homepage shows the layout of the whole room, and the overall home's energy usage and some basic controls for the house.

Interactive 3D Model.

Energy usage is shown as a yellow ring, with the limit being shown by the red square.

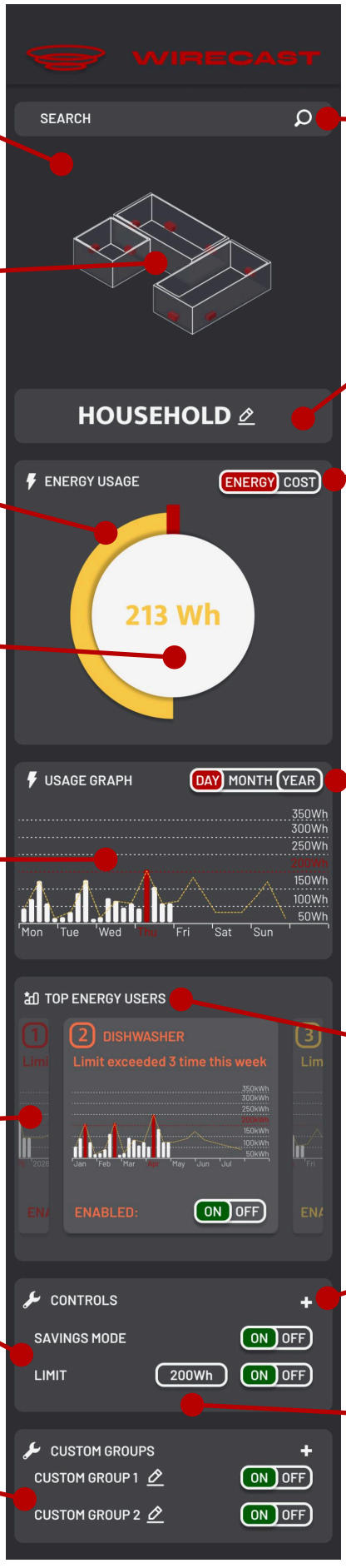
Pressing on it changes the limit from on to off, representing it as an orange ring for the extra energy.

The graph represent energy usage over time, with the limit shown as the red line.

You can look through them by dragging them as a carousel.

Key controls are shown here.

Key groups are shown here.



A search bar allows the user to quickly locate specific plugs or rooms.

The name of the household can be edited by pressing on it.

Toggling the icon changes from energy to cost for the whole page.

Toggling the icon changes from day to month to year for the whole page.

Top 3 energy users are only shown on this page, with a graph showing how much energy they've used, how often they exceed a limit, and a button to quickly enable or disable them.

Pressing plus brings up an overlay allowing you to add a control.

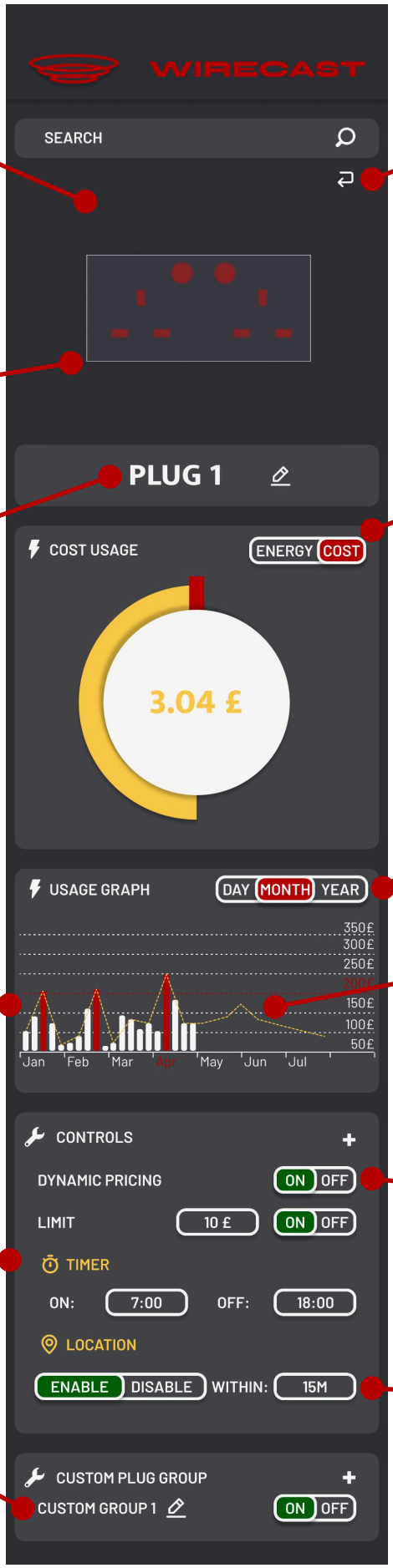
Limit can be changed by pressing on the number.

ROOM/PLUG PAGES

The room and plug pages are very similar. With information like top energy users having been removed, and lower limits and usage shown compared to the household.

While the plug model is still interactive, clicking on it does not lead to anywhere.

Name can be edited as well.



A back arrow has been added to move from plug to room, and from room to household.

Cost is selected so the whole page is shown in pounds.

Month is selected so the whole page is shown in that scale.

Red bars show weeks that went over the limit.

The yellow line shows future prediction for cost.

New command "Timer" allows a user to set a time for the product to turn on, and a time for it to turn off. Better control can be given by placing multiple timers.

Dynamic pricing is on, meaning that for this plug, energy is only supplied on off peak times when the cost is lower.

Location checks how far you're from a plug and allows you to keep it enabled or disabled when within that distance.

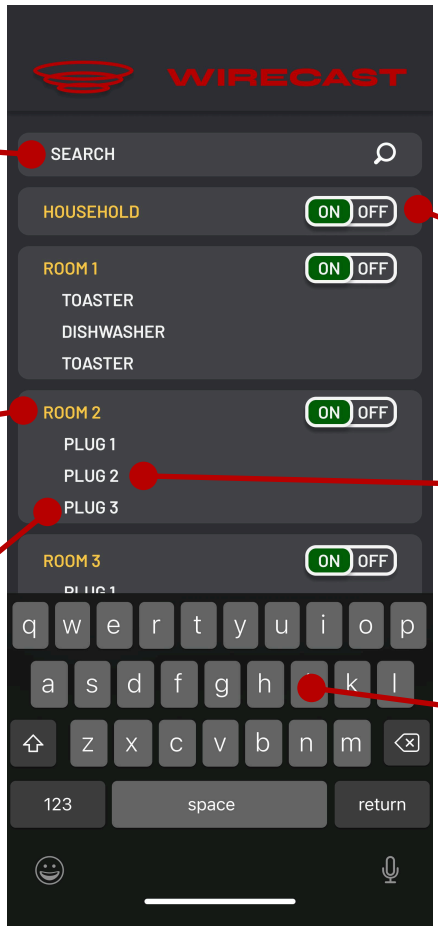
Shows what custom group it's currently part of.

OVERLAYS

The search bar brings up an overlay with the hierarchy of the household, rooms, plugs and custom groups to enable quick search of the required products.

Group names are highlighted in yellow for better visibility.

Clicking on any of them brings you to the respective page.



You can quickly toggle any of the groups on or off.

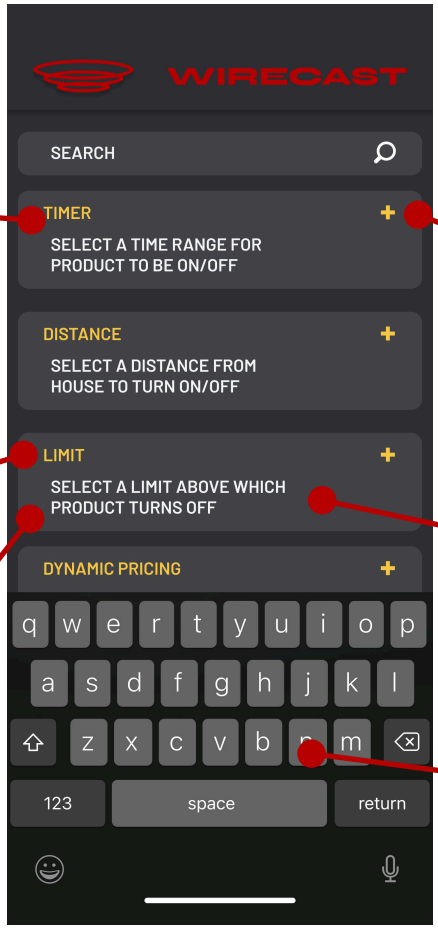
For improved visibility, plugs do not have an on/off switch but instead need to be selected for interaction.

To find a specific plug quicker, you can type in the keyboard the name.

Pressing plus next to controls brings up an overlay with the different available controls for that household, plug or room.

Control names are highlighted in yellow for better visibility.

A brief description of each control can be found underneath them.



Pressing plus allows you to add that specific control

You can then change their setting once they've been added.

To find a specific control quicker, you can type in the keyboard the name.

The Figma prototype can be tested at [17]: **https://www.figma.com/proto/yJ5vtHpYkswb2rRIV2K0yT/Francesco?page-id=0%3A1&node-id=94-30&p=f&viewport=-474%2C-1192%2C1&t=BLpBpPtoKwlg8pqH-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=85%3A83**

SYSTEM USABILITY SCALE

Running a SUS test on the product, I determined how effective the design was for both specific tasks and generally by conducting user testing. The usual set of questions were asked to the user after they were left to experiment on the app as seen below, together with the average scoring. A score above 70 is acceptable according to Bunch Media [18].

	1	2	3	4	5
1. I think I would like to use this website frequently.				X	
2. I found the website unnecessarily complex.		X			
3. I thought the website was easy to use.					X
4. I think that I would need the support of a technical savvy person to use this website.		X			
5. I found the various functions in the website were well integrated.			X		
6. I thought there was too much inconsistency in the website.	X				
7. I would imagine that most people would learn to use this website very quickly.				X	
8. I found the website very cumbersome to use.		X			
9. I felt very confident in using the website.			X		
10. I needed to learn a lot of things before I could get going with this website.		X			

To calculate the score: the results from each participant will be averaged and calculated. For each of the odd-numbered questions, I will subtract 1 from the score. For each of the even-numbered questions, I will subtract their value from 5. Then I will add up all the points and multiply by 2.5 [18].

SUS SCORE: 80

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